

EXPERIMENT 7B

DATA VISUALIZATION USING TABLEAU

Aim:

To connect, analyze, visualize, and create interactive dashboards using Tableau for effective business intelligence and data analytics.

Sample Dataset:

Artist Streaming Dataset

Algorithm 1:

Connecting Data Source

Step 1: Open Tableau Desktop

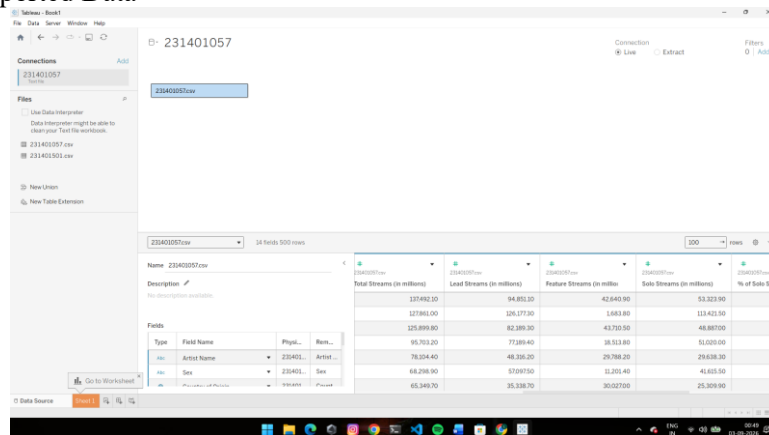
Step 2: Select Connect

Step 3: Choose Text File

Step 4: Browse Dataset

Step 5: Open Dataset

Step 6: Verify Imported Data



Algorithm 2:

Step 1: Open a new worksheet.

Step 2: Drag Primary Genre → Rows.

Step 3: Drag Total Streams (in millions) → Columns.

Step 4: Select Bar Chart visualization.

Step 5: Apply data labels.

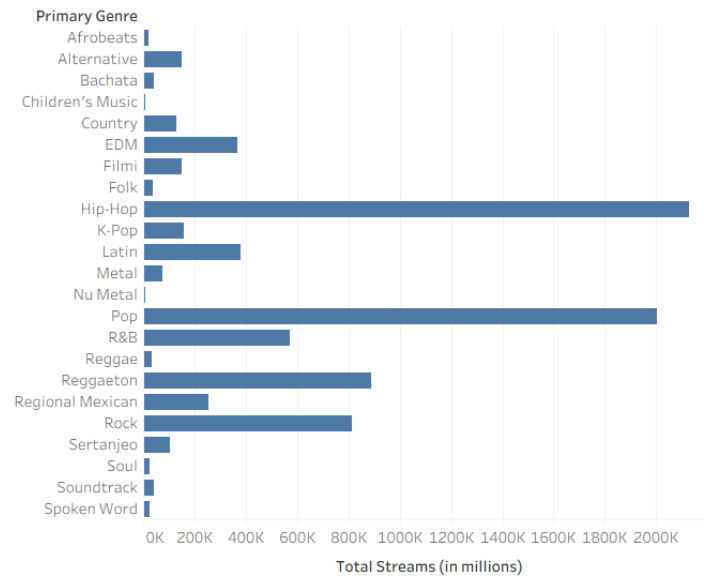
Step 6: Analyze total streams across different genres.

Output:

Total Streams by Genre bar chart.

The bar chart compares total streams across different primary genres.

Total Streams by Genre bar chart.



Algorithm 3:

Step 1: Create a new worksheet.

Step 2: Drag Primary Genre → Rows.

Step 3: Drag Total Streams (in millions) → Columns.

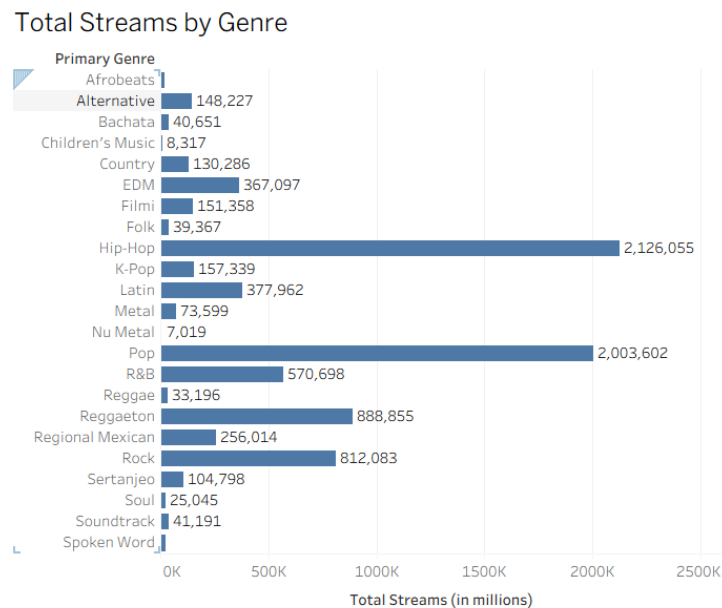
Step 4: Select Bar Chart.

Step 5: Apply data labels.

Step 6: Format the visualization.

Output:

Total Streams by Genre bar chart.



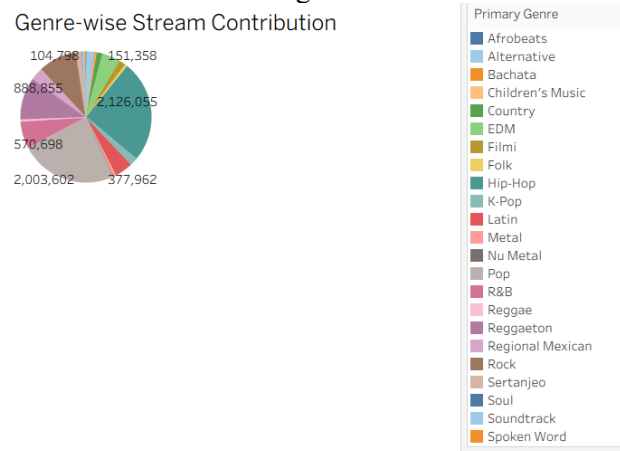
Algorithm 4:

- Step 1: Create a new worksheet.
- Step 2: Select Pie Chart.
- Step 3: Drag Primary Genre → Color.
- Step 4: Drag Total Streams (in millions) → Angle.
- Step 5: Drag Total Streams (in millions) → Label.
- Step 6: Analyze the contribution of each genre.

Output:

Genre-wise Stream Contribution pie chart.

The pie chart represents the contribution of each genre to total streams.



Algorithm 5:

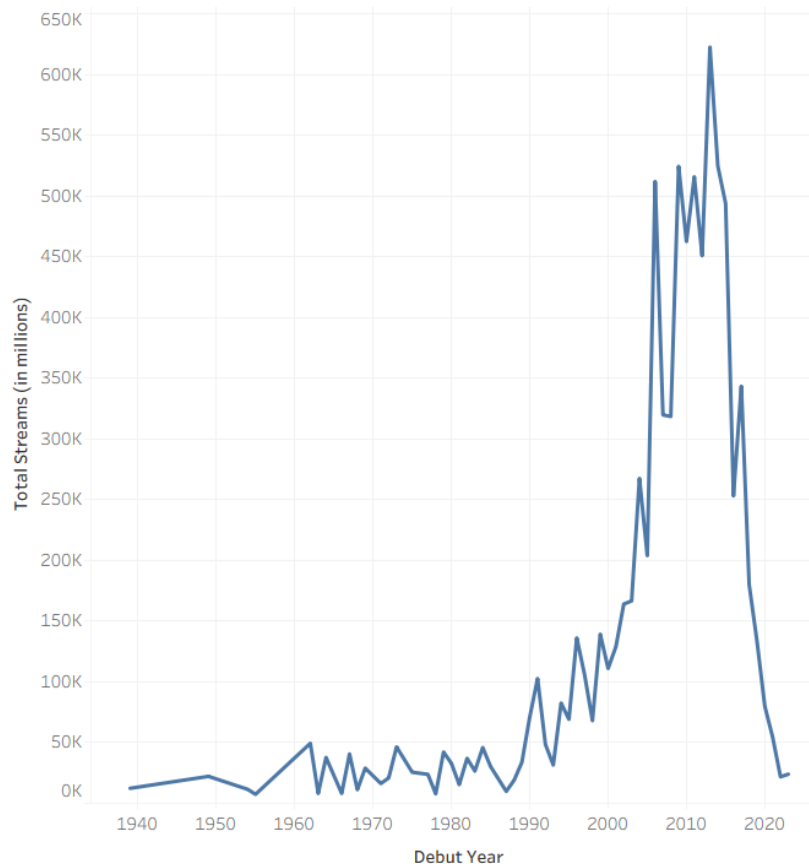
- Step 1: Create a new worksheet.
- Step 2: Drag Debut Year → Columns.
- Step 3: Drag Total Streams (in millions) → Rows.
- Step 4: Select Line Chart.
- Step 5: Apply appropriate formatting.

Output:

Total Streams Trend by Debut Year line chart.

The line chart Shows the variation in total streams over different debut years.

Total Streams Trend by Debut Year



Algorithm 6:

Creating Calculated Field

Objective

To calculate stream percentage from Total Streams.

Stream Percentage = $\text{SUM}(\text{Total Streams (in millions)}) / \text{WINDOW_SUM}(\text{SUM}(\text{Total Streams (in millions)}))$

Step 1: Select Analysis → Create Calculated Field.

Step 2: Enter the name Stream Percentage.

Step 3: Enter the formula:

$\text{SUM}([\text{Total Streams (in millions)}]) / \text{WINDOW_SUM}(\text{SUM}([\text{Total Streams (in millions)}]))$

Step 4: Click OK.

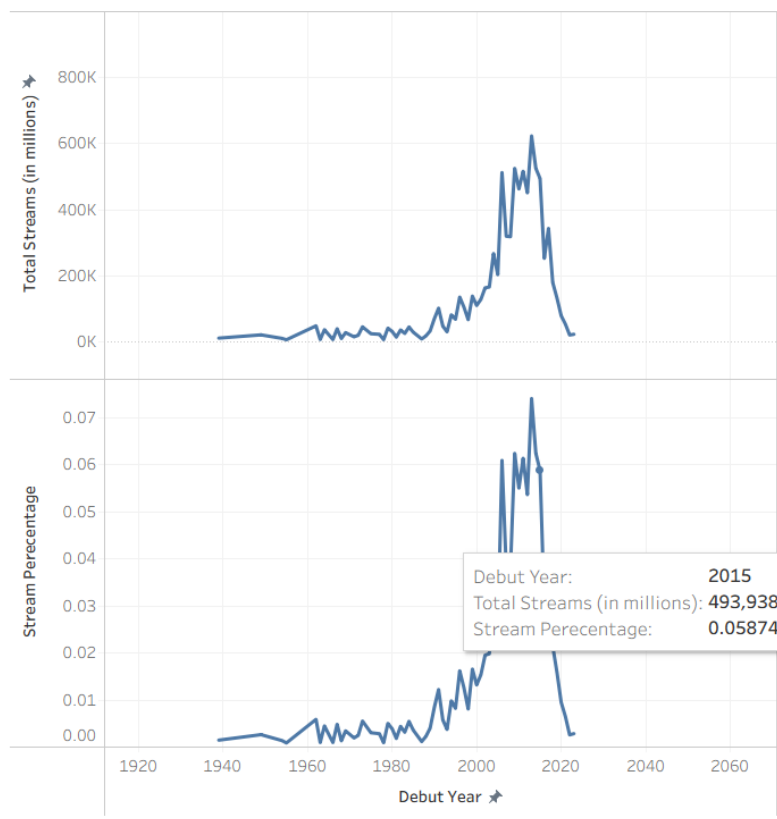
Step 5: Use Stream Percentage in the visualization.

Step 6: Format the value as required.

Output:

Stream Percentage calculated field.

Stream Percentage calculated field.

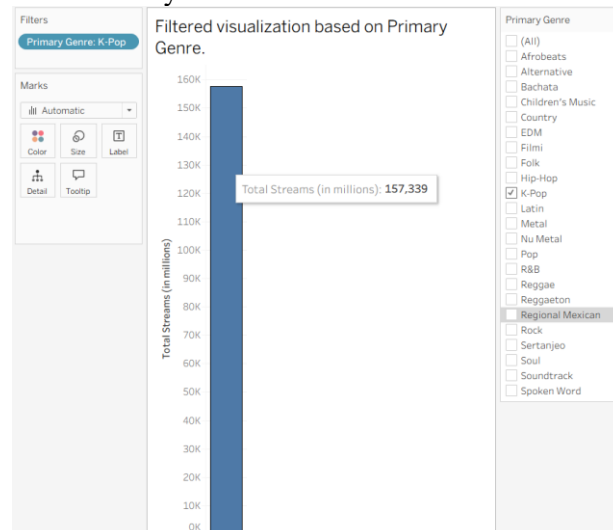


Algorithm 7:

- Step 1: Select the required worksheet.
- Step 2: Drag Primary Genre → Filters.
- Step 3: Select the required genre.
- Step 4: Apply the filter.
- Step 5: Observe the updated visualization.

Output:

Filtered visualization based on Primary Genre.



Algorithm 8:

To analyze records based on a user-defined minimum stream value.

Step 1: Select Analysis → Create Parameter.

Step 2: Name the parameter Minimum Streams.

Step 3: Set the data type and allowable range.

Step 4: Create the parameter control.

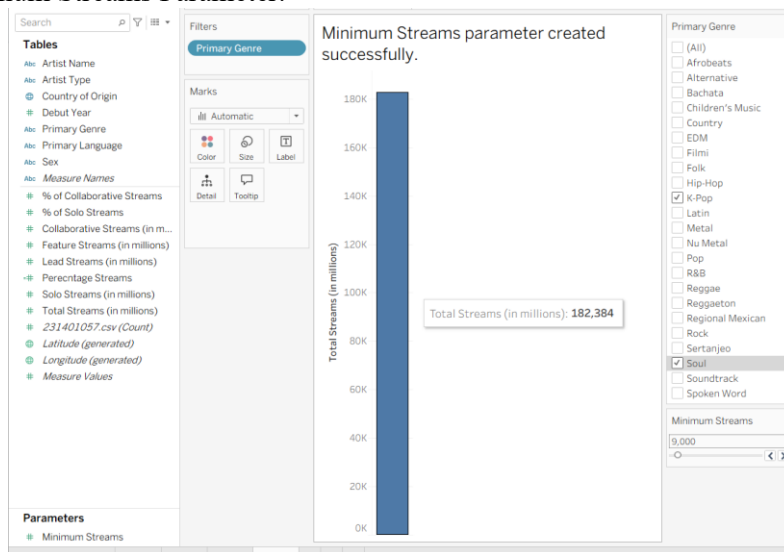
Step 5: Apply the minimum stream value.

Step 6: Display the parameter control.

Step 7: Change the minimum stream value and observe the updated visualization.

Output:

Interactive Minimum Streams Parameter.



Algorithm 9:

Step 1: Create the required worksheets.

Step 2: Select Dashboard → New Dashboard.

Step 3: Drag the worksheets onto the dashboard.

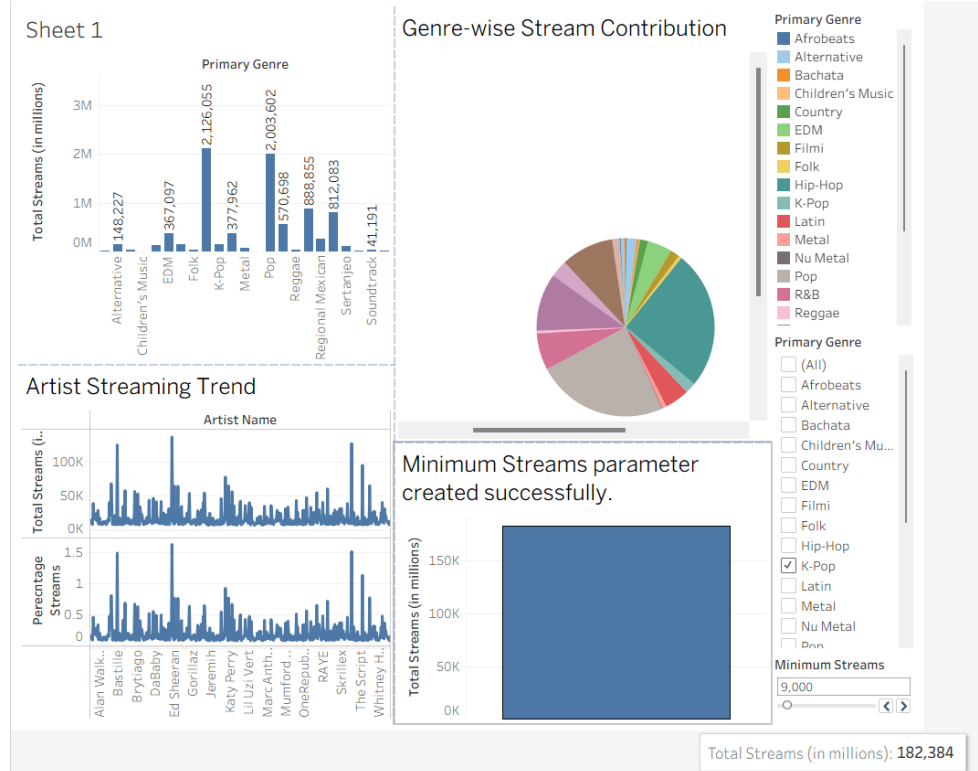
Step 4: Arrange the visualizations appropriately.

Step 5: Add the Primary Genre Filter and Minimum Streams Parameter.

Step 6: Format the dashboard.

Step 7: Save the dAashboard.

Output:



Result:

Thus, Tableau was used successfully to connect the artist streaming dataset, create worksheets, apply filters and parameters, generate visualizations, and build dashboards for data analysis.